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Big Ramsey structures in unrestricted languages and a finitely constrained counterexample

Abstract. We construct big Ramsey structures for the generic injective relational structures in languages with finitely many symbols of each positive arity, answering the existence part of Problem 6.3.2 of Balko et al. The proof combines a uniform theorem on valuation envelopes with a countable extension of the transfer and recurrence arguments of Hubička and Zucker. One embedding has uniformly bounded envelopes at every finite size. When the relation counts are unbounded, the construction uses the original valuation structure and disproves Conjecture 4.7 of Braunfeld et al. For finitely constrained classes, we construct a finite language of maximum arity three and a finite family of irreducible structures on three or four vertices whose forbidden-class limit has infinite singleton big Ramsey degree. This gives a negative instance of the existence question in Problem 6.3.3. The finite construction represents directed-graph homomorphisms by inclusion of ages of one-point realization sets; Sauer's method of persistent partitions then yields arbitrarily many unavoidable vertex colours.

Keywords: big Ramsey structure, infinite relational language, valuation envelope, free amalgamation, forbidden structure.

Ramsey's 1930 theorem asserts that a finite colouring of the k-element subsets of an infinite set becomes constant on an infinite subset [12]. For a countable relational structure K, the corresponding question is whether the subset can be required to induce a copy of K. In general, one colour no longer suffices. The big Ramsey degree of a finite structure A in K measures how many colours may be unavoidable. It asks for a bound depending on A and K, but not on the colouring or the number of colours. Determining these degrees therefore identifies precisely how much of the infinite Ramsey theorem survives the requirement that the ambient structure be preserved.

Devlin's 1979 calculation for the rationals [4] and Milliken's partition theorem for trees [11] established two features of the subject: finite tree configurations can describe unavoidable colourings, and a tree theorem can show that these are the only obstructions. For the Rado graph, Sauer proved finite upper bounds and Laflamme, Sauer and Vuksanovic identified optimal canonical partitions [10,14]. Here a canonical partition is a fixed finite partition of the copies of A such that every finite colouring becomes constant on its parts in a suitable

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copy of K, while every part occurs in every copy. Thus the types used in an upper-bound proof have structural meaning as well as a counting role.

Zucker's 2019 theory of big Ramsey structures asks that these descriptions hold simultaneously for all finite A [16]. Such a structure is an expansion of K in which the induced finite expansions record the optimal unavoidable distinctions. The expansion types must agree under restriction to smaller structures. Finiteness of the degrees alone does not provide this compatibility. Big Ramsey structures also connect these partition problems with completion flows of automorphism groups; see [9, 16]. In this paper the relevant point is combinatorial: one expansion should account for all finite colourings, rather than supply unrelated optimal partitions for individual A.

The development beyond unrestricted binary structures brought the effect of forbidden configurations into focus. Dobrinen obtained finite big Ramsey degrees for the Henson graphs [5, 6]. Zucker extended finiteness to finitely constrained binary free amalgamation classes [17], and Balko, Chodounský, Dobrinen, Hubička, Konečný, Vena and Zucker subsequently characterized their exact degrees and constructed big Ramsey structures [1].

The generic 3-uniform hypergraph provides the first test case beyond binary languages in that programme. Balko, Chodounský, Hubička, Konečný and Vena proved that it has finite big Ramsey degrees by coordinating two valuation trees [2]. Problem 6.3.1 of [1, p. 2144] asks for the next step: a characterization of the exact degrees and a big Ramsey structure realizing them simultaneously. Hubička, Konečný and Zucker have since announced a characterization by hypergraph diaries, with Ramsey and unavoidability theorems that also yield a big Ramsey structure [8, Theorems 1.4, 3.2 and 3.3].

Balko et al. formulate two natural generalizations [1, Section 6.3]. Problem 6.3.2 asks for the corresponding degree characterization and big Ramsey structure for unrestricted relational structures in languages with finitely many symbols of each arity; both the language and its arities may be unbounded. Problem 6.3.3 asks the same questions when the language is finite and finitely many irreducible structures are forbidden. These two directions organize the present paper: we establish the existence assertion in the first setting and give a negative instance of that assertion in the second.

0.1. Unrestricted structures

For the unrestricted direction, Braunfeld, Chodounský, de Rancourt, Hubička, Kawach and Konečný extended the valuation construction to languages with finitely many relations of each arity, proving finite big Ramsey degrees throughout this range [3]. Our existence result concerns the generic limit of all finite injective L-structures in each such language.

Except in Theorem 0.1 and its proof, languages have positive arities and relations hold only on tuples of distinct entries. We call this the *injective* convention; symmetry is not assumed. As in [3], the convention is essential for unbounded arity: repetitions can encode infinitely many independent relations on a fixed finite set and prevent a countable universal structure. With finitely many symbols per arity, the injective convention leaves only finitely many structures of each finite size.

For the existence argument we use a countable extension of the upper-bound transfer and finite-palette recurrence arguments of Hubička–Zucker [9, Theorem 3.10, Fact 3.4 and the proof of Theorem 3.18]. We record the form needed when the initial pullback is precompact and give its compatibility argument in Section 5. For an expansion X of K and a finite $A \in \mathrm{Age}(K)$, write $\mathcal{P}_A(X) = \{X \cdot a : a \in \mathrm{Emb}(A, K)\}$, where $X \cdot a$ is the pullback to the fixed underlying set of A. The expansion is precompact if every $\mathcal{P}_A(X)$ is finite.

Theorem 0.1 (Transfer through a precompact expansion). *Let K and M be countably infinite structures in the same relational language, and let N be a relational expansion of M. Suppose there are embeddings $\phi : K \to M$ and $\theta : M \to K$ such that $X = N \cdot \phi$ is precompact and $T_{\mathrm{emb}}(B, N) = 1$ for every finite $B \in \mathrm{Age}(X)$. Then K admits a big Ramsey structure. For every finite $A \in \mathrm{Age}(K)$,*

$$T_{\mathrm{emb}}(A, K) = \min_{\eta \in \mathrm{Emb}(K)} |\mathcal{P}_A(X \cdot \eta)|. \tag{0.1}$$

If $\mathrm{Age}(K)$ has only finitely many isomorphism types of each finite size, the expansion can be obtained by adding finitely many injective relation symbols of each arity.

No homogeneity assumption is required of K or M. The theorem and its proof allow arbitrary arities and repeated entries in the original relations; only the added relations are required to be injective. Applying it to the valuation structures constructed below gives the following result.

Corollary 0.2. *Let L be a relational language with finitely many symbols of each positive arity, and let K be the Fraïssé limit of all finite injective L-structures. Then K admits a big Ramsey structure obtained by adding finitely many injective relation symbols of each arity.*

The corollary answers the existence part of Problem 6.3.2 under this convention. Proposition 5.2 also gives an implicit minimum formula using a fixed normal-form colouring. Determining which collections of normal forms realize the minimum is a further question. The diary announcement also discusses extensions to finite relational languages [8, Section 4]. Corollary 0.2 includes unbounded arity; its minimum formula is an existence description, rather than an effective diary enumeration.

The combinatorial ingredient is a uniform envelope theorem for the valuation hypergraphs of [3, Definition 3.1]. Suppose that L has $n_r < \infty$ relation symbols of arity $r \geq 2$. An L-hypergraph has symmetric injective relations, and each finite set supports at most one relation. Put $\sigma_j = n_{j+1} + 2$, and let G_σ be the valuation hypergraph recalled in Section 3.

Theorem 0.3. *Every L-hypergraph H enumerated by ω admits an embedding $\phi : H \to G_\sigma$ preserving the enumeration such that, for every $k \geq 1$ and every $S \in [H]^k$, the set $\phi[S]$ has a valuation envelope of height at most*

$$R(k) = 2k + (k4^k + 1)^2. \tag{0.2}$$

The same embedding works for all k, and the bound is independent of σ and H.

The quantifier order matters. The earlier upper-bound argument chooses an embedding after the size k has been fixed. When the numbers n_r are unbounded, Conjecture 4.7 of [3] asserts that every single embedding has some fixed finite size for which envelope heights are unbounded. Theorem 0.3 uses the same ambient G_σ and the same notion of envelope, and gives the opposite quantifier pattern.

Corollary 0.4. *Conjecture 4.7 of* [3] *is false.*

The construction allows slices of order m to meet at an original vertex n when $n < m$. For a k-element set S, the original vertices subsequently added by closure therefore lie in

$$S \cup \{0, \dots, k-1\}.$$

This finite set controls all later closure operations. Coordinated tree embeddings preserve the resulting finite types, and the finite-dimensional form of Milliken's theorem supplies their Ramsey property. To obtain one expansion, we construct finite-stage expansions on the same underlying set and prove that each extends its predecessor. Theorem 0.1 isolates this compatibility argument from the envelope bound.

0.2. Finitely many forbidden structures

For a finite relational language L and a finite family $\mathcal{F}$ of finite irreducible L-structures, let $\mathrm{Forb}_e(\mathcal{F})$ be the class of finite injective L-structures omitting induced copies of members of $\mathcal{F}$. Irreducibility means that every pair of distinct vertices occurs together in some positive relation tuple. This condition ensures free amalgamation. Problem 6.3.3 of [1, p. 2145] asks for the big Ramsey degrees of the Fraïssé limits of these classes and asks whether they admit big Ramsey structures. Our second result gives a negative instance of the existence question.

Theorem 0.5. *There are a finite relational language L_* of maximum arity three and a finite family $\mathcal{F}_*$ of finite irreducible injective L_*-structures, each with three or four vertices, such that* $\mathrm{Forb}_e(\mathcal{F}_*)$ *has a countably infinite Fraïssé limit K_* and a one-point substructure δ satisfying*

$$T_{\mathrm{emb}}(\delta, K_*) = \infty.$$

In particular, K_* has no big Ramsey structure.

The language, forbidden family and singleton in this statement are fixed. For every $q \geq 2$ there is a q-colouring of the copies of δ such that every copy of K_* contains all q colours. Thus finite language and finite constraints, even with free amalgamation, do not ensure finite big Ramsey degrees once ternary relations are allowed. Theorem 0.5 does not classify the classes in Problem 6.3.3, nor does it decide the corresponding question for a specified class of hypergraphs.

The lower-bound method is Sauer's persistent-partition argument: finite antichains of ages of one-point realization sets give persistent vertex partitions [13, Lemma 5.1 and Theorem 5.1]; see also [15, Sections 1, 9 and 10]. Our finite presentation realizes arbitrarily

large antichains in one language and one forbidden class. For each finite loopless directed graph H, it provides a finite parameter set and a realized one-point type with realization set P_H, such that

$$\mathrm{Age}(K_*[P_H]) \subseteq \mathrm{Age}(K_*[P_J]) \quad \Longleftrightarrow \quad H \longrightarrow J,$$

where the arrow denotes a directed graph homomorphism. Directed cycles of distinct prime lengths give arbitrarily large antichains. Applying Sauer's persistence argument then gives infinite singleton degree. We include the short persistence proof for the distinguished singleton and prove the finite presentation and its age characterization in full.

Together the results distinguish two sources of difficulty beyond the binary theory. In the unrestricted case, unbounded arity requires uniform control of finite closures and compatibility of optimal expansions. Finite forbidden configurations can instead create an obstruction already at the level of vertices. The first construction provides the needed compatibility; the second exhibits such an obstruction explicitly.

1. Embeddings, expansions and trees

All structures are countable and relational, and embeddings preserve and reflect relations. Write $\mathrm{Emb}(A, B)$ for their embeddings and $\mathrm{Emb}(K)$ for $\mathrm{Emb}(K, K)$. An enumeration will be used for coding, but embeddings in Corollary 0.2 need not preserve it. For finite $A \in \mathrm{Age}(K)$, the embedding big Ramsey degree $T_{\mathrm{emb}}(A, K)$ is the least positive integer d such that every finite colouring of $\mathrm{Emb}(A, K)$ uses at most d colours on $g \circ \mathrm{Emb}(A, K)$ for some $g \in \mathrm{Emb}(K)$. If no finite d works, put $T_{\mathrm{emb}}(A, K) = \infty$. Write

$$\binom{K}{A} = \{B \subseteq K : K \restriction B \cong A\}.$$

The copy big Ramsey degree $T_{\mathrm{copy}}(A, K)$ is defined in the same way, using finite colourings of $\binom{K}{A}$ and their restrictions to $\binom{g[K]}{A}$; again the value is ∞ when no finite bound exists. We use the standard conversion between the two conventions [9, Section 2]:

$$T_{\mathrm{emb}}(A, K) = |\mathrm{Aut}(A)|\, T_{\mathrm{copy}}(A, K). \tag{1.1}$$

For singletons the two degrees therefore agree.

If X expands K and $e : A \to K$ is an embedding, let $X \cdot e$ be its pullback to the fixed underlying set of A. Put

$$\mathcal{P}_A(X) = \{X \cdot e : e \in \mathrm{Emb}(A, K)\}.$$

The expansion is *precompact* if these sets are finite, and *unavoidable* if $\mathcal{P}_A(X \cdot g) = \mathcal{P}_A(X)$ for every A and $g \in \mathrm{Emb}(K)$. It is a *big Ramsey structure* if it is unavoidable and $|\mathcal{P}_A(X)| = T_{\mathrm{emb}}(A, K) < \infty$ for every A. This is the embedding convention of [9, Definition 3.5], equivalent to the usual formulation with expansion isomorphism types and automorphism factors [16].

We say that a countable structure N has the *infinite Ramsey property* if every finite colouring of $\mathrm{Emb}(B, N)$, with $B \in \mathrm{Age}(N)$ finite, is constant on $z \circ \mathrm{Emb}(B, N)$ for some $z \in \mathrm{Emb}(N)$. An expansion F of K is *recurrent* if every $g \in \mathrm{Emb}(K)$ admits $h \in \mathrm{Emb}(K)$ such that $gh \in \mathrm{Emb}(F)$ [9, Definition 3.3(4)]. These definitions will also be used for finite-stage expansions. The existence and extension properties of Fraïssé limits are used in their standard form [7].

A strong subtree of a rooted tree is a rooted, balanced, meet-closed subtree whose levels lie on ambient levels and which, above every nonterminal node, has exactly one next-level node in each immediate ambient successor cone. A strong vector subtree has a common level set in all coordinates. Let $\mathrm{Str}_h(T_0, \dots, T_{d-1})$ denote the strong vector subtrees of height h. Height counts levels of the subtree. We use the following standard form of Milliken's theorem; see [11] and [3, Theorem 2.2].

Theorem 1.1 (Milliken). *Let $d, h \geq 1$ be integers, and let $T_0, \dots, T_{d-1}$ be rooted, finitely branching trees with no terminal nodes. Every finite colouring of $\mathrm{Str}_h(T_0, \dots, T_{d-1})$ is constant on $\mathrm{Str}_h(S_0, \dots, S_{d-1})$ for some infinite strong vector subtree $(S_i)_{i<d}$.*

All applications below have finite dimension.

2. Finite closures in valuation trees

Fix a sequence $\sigma = (\sigma_j)_{j \geq 1}$ of positive finite integers. For $i, n < \omega$, let $T_i(n)$ consist of the height-tagged functions on nonempty decreasing tuples below n such that

$$f(x_0, \dots, x_{r-1}) < \sigma_{i+r}.$$

Set $T_i = \bigcup_n T_i(n)$, ordered by restriction, and write $|f| = n$ for $f \in T_i(n)$. These are the valuation trees of [3, Section 3]. For $\ell < |f|$, its full slice is

$$f_\ell \in T_{i+1}(\ell), \qquad f_\ell(\bar{y}) = f(\ell, \bar{y}).$$

The scalar $f(\ell)$ is separate. A successor of $f \in T_i(n)$ is uniquely determined by this scalar at n and its full n-slice in $T_{i+1}(n)$. The trees have finite levels, since

$$|T_i(n)| = \prod_{r=1}^{n} \sigma_{i+r}^{\binom{n}{r}},$$

and have no terminal nodes.

Definition 2.1. For a nonempty finite $X \subseteq T_0$, let $C(X) = (C_i(X))_{i<\omega}$ be the least family containing X in coordinate zero and closed under same-coordinate meets, restrictions to all retained levels, and full slices at all retained smaller levels. Here a level is retained if it is the height of a node in any coordinate. Write $L(X)$ for the retained level set. No zero nodes or other constants are adjoined.

For $L(X) = \{\ell_0 < \cdots < \ell_{h-1}\}$ define the compression of $f \in C_i(X)$, $|f| = \ell_j$, by

$$\widehat{f}(a_1, \ldots, a_r) = f(\ell_{a_1}, \ldots, \ell_{a_r}), \qquad j > a_1 > \cdots > a_r. \tag{2.1}$$

Lemma 2.2. *The closure $C(X)$ is finite. Compression identifies its labelled diagram of nodes and ordered retained heights, including meets, restrictions, full slices and scalar values at retained levels, with $C(\widehat{X})$. Its level set becomes $\{0, \ldots, h-1\}$, and*

$$f \in C_i(X), \quad |f| = \ell_j \quad \Longrightarrow \quad i + j < h. \tag{2.2}$$

Compression is idempotent and commutes with passing to a subtuple, when the subtuple is compressed again using its own closure.

Proof. Put $N = \max\{|f| : f \in X\}$. Induction on a derivation in $C(X)$ gives

$$f \in C_i(X) \implies i + |f| \le N \quad \text{and} \quad |L(X) \cap (|f|, \infty)| \ge i. \tag{2.3}$$

For the second assertion, a slice adds its parent's height to the higher levels; a meet or restriction retains the higher levels of either parent. The first assertion gives finiteness, and the second gives (2.2).

Let $\rho(\ell_j) = j$. Directly from (2.1),

$$\widehat{f \restriction \ell_j} = \widehat{f} \restriction j, \qquad \widehat{f_{\ell_j}} = (\widehat{f})_j, \qquad \widehat{f}(j) = f(\ell_j)$$

whenever the expressions are defined. We prove injectivity and preservation of meets simultaneously by induction on the maximum ambient height of two nodes. The comparable case follows from restrictions. Otherwise, put $c = |f \wedge g|$. Since $c \in L(X)$, there are two cases:

$$\begin{aligned} f(c) \neq g(c) \quad &\Longrightarrow \widehat{f}(\rho(c)) \neq \widehat{g}(\rho(c)), \\ f_c \neq g_c \quad &\Longrightarrow \widehat{f_c} \neq \widehat{g_c} \Longrightarrow (\widehat{f})_{\rho(c)} \neq (\widehat{g})_{\rho(c)}. \end{aligned}$$

The second line uses induction, since $|f_c| = |g_c| = c$. There is no earlier disagreement, and hence

$$\widehat{f} \wedge \widehat{g} = \widehat{f \wedge g}.$$

Thus the compressed image is closed under all generating operations. Minimality gives $C(\widehat{X}) \subseteq \widehat{C(X)}$; induction on derivations gives the reverse inclusion. The same argument applied to $Y \subseteq X$ proves compatibility with subtuples. Compression on consecutive levels is the identity. ∎

For a finite nonempty label set U, the *normal form* $\mathrm{NF}(\bar{a})$ of an ordered tuple of distinct vertices in $U \times T_0$ records its labels and the compressed projections, in the original order. Distinct vertices may have the same projection. We name every such normal form by an injective relation symbol of the corresponding arity. Denote this countable expansion of $U \times T_0$ by $N_{\sigma,U}$. Lemma 2.2 implies that the normal form of a tuple determines that of each ordered subtuple.

2.1. Coordinated embeddings

Let $(S_i)_{i<\omega}$ be strong subtrees of $(T_i)_{i<\omega}$ with common level sequence $e(0) < e(1) < \cdots$. There are maps $\Psi_i : T_i \to S_i$ defined by the simultaneous recursion of [3, Lemma 3.3]. Map the root to the root. If $f \in T_i(n+1)$, the three data

$$\Psi_i(f \restriction n), \qquad \Psi_{i+1}(f_n), \qquad f(n)$$

determine an immediate ambient successor of $\Psi_i(f \restriction n)$. Define $\Psi_i(f)$ to be the unique next-level node of S_i in that cone. The recursion is well founded because both required map values have source height n. For a strong vector subtree of height h in the first h coordinates, the same definition applies on $T_i(< h - i)$.

Lemma 2.3. *The maps Ψ_i preserve meets and satisfy*

$$|\Psi_i(f)| = e(|f|), \qquad \Psi_i(f) \restriction e(m) = \Psi_i(f \restriction m), \tag{2.4}$$

$$(\Psi_i(f))_{e(m)} = \Psi_{i+1}(f_m), \qquad \Psi_i(f)(e(m)) = f(m). \tag{2.5}$$

The restriction identity allows $m \le |f|$; the slice and scalar identities require $m < |f|$. For every finite Y in the top domain,

$$C_i(\Psi_0[Y]) = \Psi_i[C_i(Y)], \qquad L(\Psi_0[Y]) = e[L(Y)]. \tag{2.6}$$

Consequently $(u, f) \mapsto (u, \Psi_0(f))$ preserves all normal forms.

Proof. Induct on the source height, simultaneously for all i. The prescribed successor cone gives (2.4)–(2.5); subsequent extensions preserve the entire slice, including its values off the selected level set. If $|f \wedge g| = c$ and f, g are incomparable, then

$$f(c) \neq g(c) \quad \text{or} \quad f_c \neq g_c.$$

In either case induction places $\Psi_i(f)$ and $\Psi_i(g)$ in different immediate cones above $\Psi_i(f \restriction c)$. Therefore

$$\Psi_i(f) \wedge \Psi_i(g) = \Psi_i(f \wedge g).$$

The generating operations commute with Ψ, so minimality and induction on derivations give (2.6). For finite vectors, (2.2) places every argument in its stated domain. Finally, repeated use of (2.5) gives

$$\Psi_i(f)(e(a_1), \ldots, e(a_r)) = f(a_1, \ldots, a_r),$$

so compression, labels and normal forms are preserved. ∎

2.2. *Finite completion and the Ramsey property*

For a finite strong vector subtree $V = (V_i)_{i<h}$ of height h, the top image $\Psi_0^V[T_0(<h)]$ is precisely $\mathrm{val}(V_0, \dots, V_{h-1})$ in [3, Definition 3.3]. Indeed, its successors are specified by a node in the preceding top level, a node in the corresponding level of the tail valuation tree, and a scalar. This is the defining recursion there. We call this image a valuation tree.

The following relative completion uses the envelope reconstruction of [3, Lemma 4.4], originating in the two-tree argument of [2, Lemma 3.3]. The closure and covariance identities verify the hypotheses needed here.

Lemma 2.4 (Relative finite completion). *Let Ψ arise from an infinite strong vector subtree $(S_i)_{i<\omega}$, and let X be a nonempty finite subset of its top image. Put $h = |L(X)|$. There is $V \in \mathrm{Str}_h(S_0, \dots, S_{h-1})$, also strong in the ambient trees, with ambient level set $L(X)$, such that*

$$\Psi_i^V(\widehat{f}) = f \qquad (f \in C_i(X)). \tag{2.7}$$

In particular $X \subseteq \mathrm{val}(V)$. The conclusion also holds for an arbitrary finite $X \subseteq T_0$, by taking $S_i = T_i$.

Proof. Write $L(X) = \{\ell_0 < \cdots < \ell_{h-1}\}$. By (2.6), $C_i(X) \subseteq S_i$. Every nonempty $C_i(X)$ has a unique node at ℓ_0: existence follows from restriction, and two such nodes would have a meet below ℓ_0.

For each $i < h$, complete $C_i(X)$ to V_i on these h levels. At an unprescribed root choose any node. At consecutive selected levels $\ell_j < \ell_{j+1}$ retain a prescribed successor when present and fill every remaining immediate S_i-cone arbitrarily. There cannot be two prescribed successors in one cone, since their meet would satisfy

$$\ell_j < |u \wedge v| < \ell_{j+1}, \qquad |u \wedge v| \in L(X).$$

Every prescribed descendant has its earlier restrictions in $C_i(X)$. Thus the construction retains $C_i(X)$ and gives a strong subtree of S_i; it is also strong in T_i. This is the completion argument of [3, Observation 2.1].

It remains to prove (2.7). Induct on j, simultaneously for $i + j < h$. The assertion is immediate at $j = 0$. For $f \in C_i(X)$ of height ℓ_j, $j > 0$, put

$$p = f \restriction \ell_{j-1}, \qquad u = f_{\ell_{j-1}}.$$

Then $p \in C_i(X)$, $u \in C_{i+1}(X)$, and induction gives

$$\Psi_i^V(\widehat{p}) = p, \qquad \Psi_{i+1}^V(\widehat{u}) = u, \qquad \widehat{f}(j-1) = f(\ell_{j-1}).$$

These are precisely the predecessor, full slice and scalar specifying the cone of f. Its unique next V_i-node is f, proving the induction step. ∎

Proposition 2.5. *For every finite nonempty U, the structure $N_{\sigma,U}$ has the infinite Ramsey property. Every finite member of its age is rigid.*

Proof. We use the Milliken reduction of [3, proof of Proposition 4.1].

Let $B = \{b_0, \dots, b_{k-1}\} \in \mathrm{Age}(N_{\sigma,U})$ be nonempty, and let $c : \mathrm{Emb}(B, N_{\sigma,U}) \to r$. Fix a realization of B with normal form τ, compressed tuple $P_\tau = ((u_j, f_j))_{j<k}$ and closure height h. The relation naming τ determines every ordered subtuple type, so an embedding of B is precisely a realization of τ. Define

$$\widetilde{c}(V) = c\big(b_j \mapsto (u_j, \Psi_0^V(f_j))\big), \qquad V \in \mathrm{Str}_h(T_0, \dots, T_{h-1}).$$

This is well defined by Lemma 2.3. Milliken's theorem gives $(S_i)_{i<h}$ on which $\widetilde{c}$ is constant. Extend the vector to all coordinates with the same level set and let

$$z(u, f) = (u, \Psi_0(f)).$$

Then $z \in \mathrm{Emb}(N_{\sigma,U})$. Let $a : B \to N_{\sigma,U}$ be an embedding. By Lemma 2.4 there is

$$V \in \mathrm{Str}_h(S_0, \dots, S_{h-1})$$

such that

$$za(b_j) = (u_j, \Psi_0^V(f_j)) \quad (j < k).$$

Consequently $c(za) = \widetilde{c}(V)$ is constant.

Every finite member is rigid: the normal form of a tuple listing all its vertices records an ordered list of pairwise distinct labelled compressed nodes, by Lemma 2.2. The empty case is immediate. ∎

3. A uniform embedding into the valuation hypergraph

We now assume $\sigma_j = n_{j+1} + 2 \geq 2$. In G_σ, a set $\{f_0, \dots, f_{r-1}\}$ of nodes of strictly decreasing heights has colour $s \in \{1, \dots, n_r\}$ exactly when

$$f_0(|f_1|, \dots, |f_{r-1}|) = s.$$

The values 0 and $\sigma_{r-1} - 1$ both represent no relation, as in [3, Definition 3.1]. Sets with height ties have no relation. An envelope of a finite set is a valuation tree containing it, with no requirement of minimality [3, Definition 4.3]. If the language has a largest arity, we keep the unused alphabet sizes equal to 2, whereas Section 4 of that paper sets them to 1. For unbounded arity, and in particular under the hypothesis of Conjecture 4.7, our signature agrees exactly with theirs.

Enumerate H by ω. For $v > z_0 > \cdots > z_{t-1}$ let $c(v, \bar{z}) \in \{0, \dots, \sigma_t - 2\}$ be its genuine colour. For every nonempty decreasing tuple $\bar{y}$ put

$$D(\bar{y}) = \max_{0 \leq d \leq y_0} (\sigma_{d+|\bar{y}|} - 2). \tag{3.1}$$

Introduce distinct markers $b_s(\bar{y})$ for $1 \leq s \leq D(\bar{y})$. Place those beginning with n in a finite block immediately before vertex n, with an arbitrary fixed order inside each block. Every

block is finite, so the resulting order has type ω. Identify its positions with ω, denote original positions by o_v, and put $O = \{o_v : v < \omega\}$ and $B = \omega \setminus O$. In particular $b_s(\bar{y}) < o_{y_0}$. An m-slice means successive slicing along a decreasing prefix of length m; the 0-slice is the node itself. A slice is *original* if every position in its prefix belongs to O.

Define $\phi(v) \in T_0(o_v)$ by

$$\phi(v)(o_{z_0}, \dots, o_{z_{t-1}}) = c(v, \bar{z}), \tag{3.2}$$

$$\phi(v)(o_{x_0}, \dots, o_{x_{m-1}}, b_s(\bar{y})) = \sigma_{m+1} - 1 \tag{3.3}$$

in the second line precisely when $(v) \frown \bar{x} \frown \bar{y}$ is strictly decreasing, $m = |\bar{x}| \le y_0$, and $c(v, \bar{x} \frown \bar{y}) = s > 0$. The prefix $\bar{x}$ may be empty. All other entries are zero. Equation (3.1) ensures that every required marker exists. The prescribed value in (3.3) lies in the correct alphabet. Equation (3.2) proves that ϕ is an embedding, and its strictly increasing heights preserve the enumeration.

This modifies the fixed-depth marker construction of [3, Lemma 4.3]. The cutoff now depends on the first vertex of the suffix, rather than on a chosen size of finite substructure.

Lemma 3.1. *Every slice of an image node whose prefix contains a position in B is zero. Suppose that $f, g \in T_m$ are incomparable and each is either an original m-slice of an image node or a zero node. If $|f \wedge g| = o_n$, then $n < m$.*

Proof. A nonzero entry containing a marker has exactly one marker, in its last position. If a slice prefix contains a marker, every entry of that slice therefore vanishes.

Let $f, g \in T_m$ be the nodes in the second assertion; allow one to be zero. Suppose, towards a contradiction, that

$$|f \wedge g| = o_n, \qquad n \ge m.$$

Then $f \restriction o_n = g \restriction o_n$, but there is a distinguishing input whose largest coordinate is o_n. Each nonzero comparator has an original outer prefix of length m, all of whose vertices exceed n.

Case 1. The input is $(o_{z_0}, \dots, o_{z_{t-1}})$, where $z_0 = n$. Choose a genuine colour $s > 0$ taken on precisely one side. Since $m \le n$, (3.1) supplies the marker $b = b_s(\bar{z})$.

Case 2. The input contains a marker. A nonzero value requires the form

$$(o_{z_0}, \dots, o_{z_{q-1}}, b_s(\bar{y})), \qquad q \ge 1, \quad z_0 = n, \quad m + q \le y_0.$$

The tuple $\bar{z} \frown \bar{y}$ is decreasing. For two original comparators these admissibility conditions coincide; hence exactly one corresponding genuine colour with suffix $\bar{z} \frown \bar{y}$ equals s. The same conclusion holds against zero. Since $m \le n$, the marker $b = b_s(\bar{z} \frown \bar{y})$ exists.

In either case $b < o_n$, and (3.3) gives

$$\{f(b), g(b)\} = \{0, \sigma_{m+1} - 1\},$$

contrary to $f \restriction o_n = g \restriction o_n$. These cases exhaust the possible nonzero entries. ∎

Lemma 3.2. *For every $S \in [H]^k$, $k \geq 1$, the intrinsic closure of $\phi[S]$ has at most $R(k)$ levels.*

Proof. The finite container below replaces the closure construction of [3, Lemma 4.4]. Put $A = S \cup \{0, \dots, k-1\}$ and $N = \max o[S]$. Then $|A| \leq 2k$ and $\max A = \max S$. Let F_i be the family of original slices

$$\phi(v)_{o_{a_1},\dots,o_{a_i}}, \qquad v \in S, \quad v > a_1 > \cdots > a_i, \quad a_j \in A.$$

Thus $F_0 = \phi[S]$, $F_i = \varnothing$ for $i \geq |A|$, and

$$P = \sum_i |F_i| \leq k2^{|A|} \leq k4^k. \tag{3.4}$$

Write 0^i_N for the zero node at height N in coordinate i and set

$$L^* = o[A] \cup \bigcup_{i:F_i \neq \varnothing} \{|u \wedge w| : u, w \in F_i \cup \{0^i_N\}\}. \tag{3.5}$$

We claim that

$$L^* \cap O \subseteq o[A]. \tag{3.6}$$

A comparable meet has one of the original heights already in $o[A]$. For an incomparable meet $|u \wedge w| = o_n$, there are two cases. If either original prefix uses $a < k$, then $o_n < \min\{|u|, |w|\} \leq o_a < o_k$, so $n < k$. Otherwise the prefixes use only S, whence $i < k$ and Lemma 3.1 gives $n < i < k$. These statements include the comparison with 0^i_N, proving (3.6).

Put $p_i = |F_i|$. The zero–zero meet contributes $N \in o[A]$, and thus

$$|L^*| \leq 2k + \sum_i (p_i^2 + 2p_i) \leq 2k + P^2 + 2P \leq R(k). \tag{3.7}$$

Let $L^* = \{\ell_0 < \cdots < \ell_{h-1}\}$ and define, for $i < h$,

$$Q_i = \{u \restriction \ell_j : u \in F_i,\ \ell_j \leq |u|\} \cup \{0^i_{\ell_j} : i + j < h\}. \tag{3.8}$$

Every $u \in F_i$ has i higher levels in $o[A]$, namely its original vertex and all but the last slicing parameter. Consequently

$$f \in Q_i, \quad |f| = \ell_j \quad \Longrightarrow \quad i + j < h. \tag{3.9}$$

We verify closure of Q.

(1) Restrictions remain in (3.8).

(2) For $u, w \in F_i \cup \{0^i_N\}$ and admissible $a, b \in L^*$,

$$(u \restriction a) \wedge (w \restriction b) = u \restriction \min\{a, b, |u \wedge w|\} \in Q_i.$$

The minimum belongs to L^* by (3.5).

(3) Let $f \in Q_i$ have height ℓ_j, and let $t < j$. If f is zero or $\ell_t \in B$, then

$$f_{\ell_t} = 0^{i+1}_{\ell_t} \in Q_{i+1}, \qquad i + 1 + t \le i + j < h.$$

Otherwise f is a restriction of some $u \in F_i$ and $\ell_t = o_a$ for $a \in A$, by (3.6). Then $f_{\ell_t} = u_{o_a} \in F_{i+1} \subseteq Q_{i+1}$.

Thus $\phi[S] \subseteq Q_0$ and $C_i(\phi[S]) \subseteq Q_i$ for every i. In particular $L(\phi[S]) \subseteq L^*$, and (3.7) finishes the proof. ∎

Proof of Theorem 0.3 *and Corollary* 0.4. Apply Lemma 2.4 in the full ambient trees to $\phi[S]$. It gives a genuine valuation envelope of height $|L(\phi[S])| \le R(k)$. The map ϕ was chosen independently of k and S. If n_r is unbounded and H is universal, choosing a threshold greater than $R(k)$ contradicts Conjecture 4.7 for every positive k; for $k = 0$, any valuation tree of height one contains the empty set. ∎

Remark 3.3. The stronger k-enveloping condition of [3, Definition 4.2] excludes every original meet of incomparable slices. Lemma 3.1 permits $n < m$; by (3.6), these exceptions do not prevent uniformly bounded envelopes.

4. From hypergraphs to relational structures

Return to the language of Corollary 0.2. Write L_r for its r-ary symbols and let $U = \{0,1\}^{L_1}$. For $r \ge 2$, on an increasingly enumerated r-element support record the full profile

$$\big(R(a_{\pi(0)}, \dots, a_{\pi(r-1)})\big)_{R \in L_r,\, \pi \in \mathfrak{S}_r}.$$

There are $q_r = 2^{|L_r| r!}$ profiles. Identify them with $\{0, \dots, q_r - 1\}$, taking the all-false profile to zero, and put $\sigma_{r-1} = q_r + 1$. This is the usual reduction to coloured hypergraphs used in [3, Proposition 5.2].

Define an L-structure M on $U \times T_0$. Unary relations read the label. For an r-element support of distinct heights, list the vertices in increasing height order and evaluate the highest node on the other heights in decreasing order. Decode a value below q_r as its profile and decode the reserved value q_r as the all-false profile. On supports with a height tie, make every relation of arity at least two false. Relations on repeated entries are always false.

An enumeration of K makes its support profiles a coloured hypergraph. Apply the construction of Section 3 and retain the unary labels. This gives an embedding $\phi : K \to M$. It preserves the chosen enumeration by heights, so every orientation in a profile is decoded in the correct order. Every finite substructure of M is an injective L-structure, so $\mathrm{Age}(M) \subseteq \mathrm{Age}(K)$. Countable universality of the Fraïssé limit gives an embedding $\theta : M \to K$.

Expand M by the normal-form predicates of $N_{\sigma,U}$, and call the result N. Every L-relation is determined by the normal form of its tuple: that form records labels, height order and ties, and the valuation on the support. Thus adding the L-relations does not change embeddings between the relevant finite expanded structures or self-embeddings of the source. Proposition 2.5 applies to N.

Proposition 4.1. *The expansion $X = N \cdot \phi$ of K is precompact. For every k, only finitely many of its added k-ary predicates are nonempty. If A has $k \geq 1$ vertices and*

$$B_\sigma(R) = \sum_{n<R} \prod_{r=1}^{n} \sigma_r^{\binom{n}{r}},$$

then

$$|\mathcal{P}_A(X)| \leq \big(|U| B_\sigma(R(k))\big)^k. \tag{4.1}$$

Proof. Lemma 3.2 bounds the closure height of every projected k-tuple in $\phi[K]$. Compression therefore places its projected normal form in the finite set $T_0(< R(k))^k$. Including labels gives the displayed bound. By heredity of normal forms, the whole pullback expansion on the tuple is determined by this one normal form. Only finitely many such forms occur globally at each size, which also proves the assertion about nonempty predicates. The empty structure has a unique pullback type. ∎

5. Compatible optimal expansions

We prove Theorem 0.1 using the upper-bound transfer [9, Theorem 3.10], finite-palette minimization [9, Fact 3.4], and the recurrence argument in the proof of [9, Theorem 3.18]. Complete source types are first named by finitely many predicates at each stage. Recurrence then makes the stages extend one another on the same underlying set. We use only the recurrence argument of the last reference; optimality below is proved separately on the finite family handled at each stage.

Fix K, M, N, ϕ, θ as in Theorem 0.1, and put $X = N \cdot \phi$. Choose one representative of each isomorphism type of nonempty finite substructures of K, labelling each k-point representative by $\{0, \dots, k-1\}$. Take increasing finite sets $I_0 = \varnothing \subseteq I_1 \subseteq \cdots$ exhausting them and closed under nonempty substructure types. Such an exhaustion exists because K is countable and a finite structure has only finitely many induced substructures.

For each representative A, put $Q_A = \mathcal{P}_A(X)$. For $p \in Q_A$, introduce an injective $|A|$-ary predicate $R_{A,p}$. Let N_n expand M by the symbols indexed by $A \in I_n$. Writing $k = |A|$, set

$$R_{A,p}^{N_n}(f(0), \dots, f(k-1)) \quad \Longleftrightarrow \quad f \in \mathrm{Emb}(A, M) \ \text{ and } \ N \cdot f = p.$$

All other tuples give false instances. Set $Y_n = N_n \cdot \phi$, so $Y_0 = K$. Every N-self-embedding preserves these predicates; hence $\mathrm{Emb}(N) \subseteq \mathrm{Emb}(N_n)$.

For $\eta \in \mathrm{Emb}(K)$, put $F = Y_n \cdot \eta$. If $A \in I_n$ and $a, b \in \mathrm{Emb}(A, K)$, then

$$F \cdot a = F \cdot b \quad \Longleftrightarrow \quad X \cdot (\eta a) = X \cdot (\eta b). \tag{5.1}$$

Indeed, $R_{A,p}$ on the fixed enumeration records the complete source type, which in turn determines all predicates on ordered subtuples. Consequently each $p \in \mathcal{P}_A(F)$ determines a unique $A^p \in Q_A$, and

$$\mathrm{Emb}(p, N_n) = \mathrm{Emb}(A^p, N), \qquad T_{\mathrm{emb}}(p, N_n) = 1. \tag{5.2}$$

The equality follows from the distinguished positive naming predicate. The degree-one assertion follows from the hypothesis on A^P and $\mathrm{Emb}(N) \subseteq \mathrm{Emb}(N_n)$.

Lemma 5.1. *Let $n \geq 0$, $\eta \in \mathrm{Emb}(K)$ and $F = Y_n \cdot \eta$. If $\mathcal{P}_A(F)$ is unavoidable for every $A \in I_n$, then F is recurrent and $|\mathcal{P}_A(F)| = T_{\mathrm{emb}}(A, K)$ for every $A \in I_n$. For a fixed $A \in I_n$, the upper bound does not require unavoidability.*

Proof. By (5.2), the upper-bound transfer [9, Theorem 3.10], applied to $(M, \phi\eta, \theta)$ and N_n, gives

$$T_{\mathrm{emb}}(A, K) \leq \sum_{p \in \mathcal{P}_A(F)} T_{\mathrm{emb}}(p, N_n) = |\mathcal{P}_A(F)|.$$

Unavoidability gives the reverse inequality by [9, Fact 3.4(1)].

For recurrence we use the argument of [9, proof of Theorem 3.18]. Fix $g \in \mathrm{Emb}(K)$. Simultaneously homogenize the finitely many colourings

$$d_{A,p} : \mathrm{Emb}(p, N_n) \longrightarrow \mathcal{P}_A(F), \qquad d_{A,p}(f) = F \cdot (g\theta f), \quad A \in I_n, p \in \mathcal{P}_A(F),$$

using (5.2). Let $z \in \mathrm{Emb}(N_n)$ witness this and put $e = g\theta z\phi\eta$. There are maps π_A satisfying

$$F \cdot (ea) = \pi_A(F \cdot a), \qquad \pi_A[\mathcal{P}_A(F)] = \mathcal{P}_A(F \cdot e) = \mathcal{P}_A(F). \tag{5.3}$$

Thus the π_A are permutations. Choose a common multiple $q \geq 1$ of their orders. Then $F \cdot (e^q a) = F \cdot a$ for all $A \in I_n$. These types determine every added predicate; on repeated tuples and tuples of the wrong base type the predicates are false. Therefore $e^q \in \mathrm{Emb}(F)$. Since $e^q = g[(\theta z\phi\eta)e^{q-1}]$, recurrence follows. The case $I_n = \varnothing$ is immediate. ∎

Proof of Theorem 0.1. Construct $\eta_n \in \mathrm{Emb}(K)$ and $F_n = Y_n \cdot \eta_n$ so that

(1) F_n is recurrent;

(2) its palettes on I_n are optimal and unavoidable;

(3) F_n extends F_{n-1} on the same underlying set.

Start with $\eta_0 = \mathrm{id}_K$ and $F_0 = K$. Given the preceding stage, put $H_n = Y_n \cdot \eta_{n-1}$ and choose $g \in \mathrm{Emb}(K)$ minimizing

$$\sum_{A \in I_n} |\mathcal{P}_A(H_n \cdot g)|. \tag{5.4}$$

For every $u \in \mathrm{Emb}(K)$, the inclusions $\mathcal{P}_A(H_n \cdot gu) \subseteq \mathcal{P}_A(H_n \cdot g)$ are equalities by minimality. Hence these palettes are unavoidable; this is [9, Fact 3.4(2)].

By recurrence of F_{n-1} choose $h \in \mathrm{Emb}(K)$ with $gh \in \mathrm{Emb}(F_{n-1})$, and set

$$\eta_n = \eta_{n-1} gh, \qquad F_n = Y_n \cdot \eta_n.$$

The palettes remain unchanged, so Lemma 5.1 gives (1)–(2). Moreover,

$$F_n \restriction L^*_{n-1} = F_{n-1} \cdot gh = F_{n-1}, \tag{5.5}$$

where L^*_{n-1} is the language of F_{n-1}, proving (3).

Let $K^* = \bigcup_n F_n$. If $A \in I_n$, a later predicate $R_{B,p}$ has $B \notin I_n$. Such a B cannot embed into A by closure of I_n under substructure types. Every later predicate is therefore false on an A-copy, and (5.5) yields

$$K^* \cdot a = F_n \cdot a \qquad (a \in \mathrm{Emb}(A, K)), \tag{5.6}$$

after restoring the empty later predicates. Thus the final palettes are finite, optimal and unavoidable, so K^* is a big Ramsey structure. The empty structure has degree one.

By (5.1), the palette of $Y_n \cdot \eta$ on $A \in I_n$ is in bijection with $\mathcal{P}_A(X \cdot \eta)$. Lemma 5.1 gives the upper bound in (0.1) for every η, and equality holds at $\eta = \eta_n$ by optimality. This proves the minimum formula. Finally, the pairs (A, p) form a countable set. If there are finitely many base types of each finite size, only finitely many such pairs have $|A| = k$, giving finitely many added k-ary symbols. ∎

The union above is a union of compatible interpretations on K; no intersection of the ranges of η_n is taken. We do not assert recurrence of K^* or simultaneous optimality of one pullback $X \cdot \eta$ for all finite structures.

Proof of Corollary 0.2. Use M, N, ϕ, θ from Section 4. Proposition 4.1 gives precompactness of $X = N \cdot \phi$, while Proposition 2.5 gives degree one for every finite member of $\mathrm{Age}(N)$, hence of $\mathrm{Age}(X)$. Theorem 0.1 applies. For each k, only the finitely many symbols of arity at most k can be nonempty on an injective k-point structure. Thus there are finitely many base types of each size, and the theorem's language-size conclusion applies. ∎

Return now to the structures and embeddings of Section 4.

Proposition 5.2. *For finite nonempty $A \in \mathrm{Age}(K)$ with vertices labelled $0, \dots, k-1$, put $\chi_A(e) = \mathrm{NF}(\phi e(0), \dots, \phi e(k-1))$. Then*

$$T_{\mathrm{emb}}(A, K) = \min_{\eta \in \mathrm{Emb}(K)} |\chi_A[\eta \circ \mathrm{Emb}(A, K)]|. \tag{5.7}$$

Every finite colouring of $\mathrm{Emb}(A, K)$ has a self-copy on which its value depends only on the K^*-type. Moreover,

$$T_{\mathrm{copy}}(A, K) = \frac{T_{\mathrm{emb}}(A, K)}{|\mathrm{Aut}(A)|}. \tag{5.8}$$

Proof. Normal forms determine precisely the full pullback types in X, so (5.7) is (0.1). For canonization, put $d = |\mathcal{P}_A(K^*)|$ and apply the degree bound to $e \mapsto (c(e), K^* \cdot e)$. On a self-copy there are at most d pairs, and unavoidability forces all d types to occur. Each type therefore has exactly one c-colour; compare [9, Proposition 3.8]. Finally, (5.8) is the standard identity (1.1). ∎

6. A finitely constrained class with infinite singleton degree

The unrestricted classes considered above have finite big Ramsey degrees. Finite forbidden families produce a different situation. We construct one finite language in which constraints on at most four vertices force infinite degree already for a singleton. The obstruction is expressed by the ages of one-point realization sets, rather than by the sizes of coding envelopes. These ages are the ranks of types in Sauer's work on vertex partitions [15]. The passage from incomparable ages to persistent vertex colours is Sauer's persistent-partition argument [13, Section 5]. Here the construction realizes an infinite antichain of these ages in one finitely constrained class.

A finite relational structure is *irreducible* if each pair of distinct vertices occurs together in a positive relation tuple. Write $\mathrm{Forb}_e(\mathcal{F})$ for the finite structures admitting no embedding of a member of $\mathcal{F}$. In this section, embeddings preserve and reflect relations, and relations hold only on tuples of distinct vertices.

The following form also gives the conclusion for larger finite structures containing the distinguished singleton.

Theorem 6.1. *There are a finite relational language L_* of maximum arity three, a finite family $\mathcal{F}_*$ of irreducible L_*-structures, each with three or four vertices, and a one-point structure δ such that the following holds. The class* $\mathrm{Forb}_e(\mathcal{F}_*)$ *is a free amalgamation class with a countably infinite Fraïssé limit K_*, and*

$$T_{\mathrm{emb}}(\delta, K_*) = T_{\mathrm{copy}}(\delta, K_*) = \infty.$$

More generally, both degrees are infinite for every finite $A \in \mathrm{Age}(K_*)$ containing a vertex of type δ.

The language and the forbidden family in this theorem are fixed. In particular, they do not depend on the number of colours used to witness infinite degree. The theorem supplies an obstruction relevant to Problem 6.3.3 of [1]; it does not classify the finitely constrained free amalgamation classes.

6.1. Realization sets and persistent colourings

Let M be a countable homogeneous structure in a finite relational language whose age has free amalgamation. A *one-point type* over a finite $A \subseteq M$ is a structure t on $A \cup \{*\}$ extending $M \restriction A$ and belonging to $\mathrm{Age}(M)$. Its realization set and its age are

$$P(t) = \{x \in M \setminus A : \mathrm{id}_A \cup \{(*, x)\} \in \mathrm{Emb}(t, M)\},$$
$$C(t) = \mathrm{Age}(M \restriction P(t)).$$

We call a structure *age indivisible* if, in every finite partition of its vertex set, some part has the same age as the whole structure.

The following age-indivisibility result applies to types in arbitrary finite relational languages.

Lemma 6.2. *For every realized one-point type t, the structure $M \restriction P(t)$ is age indivisible. The same holds for every countable structure with age $C(t)$.*

Proof. Homogeneity makes $P(t)$ an orbit of the pointwise stabilizer of its finite parameter set. By [13, Theorem 3.2 and Lemma 4.1], free amalgamation gives age indivisibility of this orbit. The property depends only on the age by [13, Theorem 3.1], proving the second assertion. The finite-colour statement follows from the two-colour statement by induction. ∎

The next proposition is a formulation of Sauer's persistence argument [13, Lemma 5.1 and Theorem 5.1]. We retain the short proof for the specified singleton type δ.

Proposition 6.3. *For $1 \le i \le q$, let t_i be a realized one-point type over a finite set $A_i \subseteq M$. Suppose their points have the same singleton type δ and their ages $C_i = C(t_i)$ are pairwise incomparable under inclusion. There is a colouring*

$$\chi : M_\delta \longrightarrow \{1, \dots, q\}, \qquad M_\delta = \{x \in M : M \restriction \{x\} \cong \delta\},$$

such that, for every $f \in \mathrm{Emb}(M, M)$ and $i \le q$,

$$\mathrm{Age}\big(M \restriction (f[P(t_i)] \cap \chi^{-1}(i))\big) = C_i. \tag{6.1}$$

In particular, all q colours occur in every copy of M.

Proof. Fix an enumeration $\nu : M \to \omega$ and, for finite $B \subseteq M$, put

$$w(B) = \sum_{x \in B} 2^{\nu(x)}.$$

For $x \in M_\delta$, minimize $w(\alpha[A_i])$ over all $i \le q$ and all $\alpha \in \mathrm{Emb}(t_i, M)$ with $\alpha(*) = x$. Such embeddings exist by homogeneity. If two indices attain the same minimum, their boundary images coincide, since w is injective. The two pointed types are then isomorphic. Homogeneity implies that their realization sets have equal ages, contrary to incomparability. Let $\chi(x)$ be the unique minimizing index.

Fix i and f, and write $Y = f[P(t_i)]$, $B = f[A_i]$. If $x \in Y$ and $\chi(x) \neq i$, a minimizing witness has index $j \neq i$ and boundary of weight strictly less than $w(B)$. Only finitely many transported realization sets $Q_1, \dots, Q_m$ can arise from these witnesses. For each $a \le m$,

$$\mathrm{Age}(M \restriction Q_a) = C_{j_a}, \qquad j_a \neq i, \qquad D_a \in C_i \setminus C_{j_a}$$

for some finite D_a. Partition Y into

$$Z_a = (Y \cap Q_a) \setminus \bigcup_{b<a} Q_b \quad (1 \le a \le m), \qquad Z_0 = Y \setminus \bigcup_{a=1}^{m} Q_a.$$

No Z_a with $a \ge 1$ contains D_a. Since $\mathrm{Age}(M \restriction Y) = C_i$, Lemma 6.2 yields $\mathrm{Age}(M \restriction Z_0) = C_i$. By construction $\chi[Z_0] = \{i\}$, which proves (6.1). ∎

6.2. The forbidden family

We now encode the homomorphism order of finite directed graphs in the ages of realization sets. The language is

arity	symbols
1	$O, V, \mathsf{Path}, \mathsf{Test}, \mathsf{First}, \mathsf{Last}$
2	E, N, Q, G, L, R
3	S_0, S_1, S_2, U_L, U_R.

An *E-complete* set satisfies $E(x, y)$ for all distinct x, y in it. Put

$$\mathcal{T} = \{(0,0), (0,1), (1,2), (2,2)\}.$$

The following conditions describe the legal structures. Every displayed tuple of named vertices consists of distinct vertices and is E-complete. An implication is required only when its displayed hypotheses hold.

(1) On $\{a, h, v\}$, if

$$O(a) \wedge \mathsf{Path}(a) \wedge V(h) \wedge V(v) \wedge Q(h, v),$$

then

$$\sum_{s=0}^{2} \mathbf{1}_{S_s(a,h,v)} = 1, \qquad \mathsf{First}(a) \Rightarrow S_0(a, h, v), \qquad \mathsf{Last}(a) \Rightarrow S_2(a, h, v). \tag{6.2}$$

(2) On $\{a, a', h, v\}$, if

$$O(a) \wedge \mathsf{Path}(a) \wedge O(a') \wedge \mathsf{Path}(a') \wedge V(h) \wedge V(v) \wedge N(a, a') \wedge Q(h, v),$$

then

$$S_s(a, h, v) \wedge S_t(a', h, v) \;\Rightarrow\; (s, t) \in \mathcal{T}. \tag{6.3}$$

(3) On $\{b, a, h, v\}$, if

$$O(b) \wedge \mathsf{Test}(b) \wedge O(a) \wedge \mathsf{Path}(a) \wedge V(h) \wedge V(v) \wedge Q(h, v),$$

then

$$\begin{aligned} L(b, a) &\Rightarrow \big(U_L(b, h, v) \leftrightarrow S_1(a, h, v)\big), \\ R(b, a) &\Rightarrow \big(U_R(b, h, v) \leftrightarrow S_1(a, h, v)\big). \end{aligned} \tag{6.4}$$

(4) On $\{b, h, v, w\}$, if

$$O(b) \wedge \mathsf{Test}(b) \wedge V(h) \wedge V(v) \wedge V(w) \wedge Q(h, v) \wedge Q(h, w) \wedge G(v, w),$$

then

$$\neg\big(U_L(b, h, v) \wedge U_R(b, h, w)\big). \tag{6.5}$$

For each condition, include in $\mathcal{F}_*$ every E-complete injective L_*-structure on its displayed three- or four-element labelled domain that violates that condition. Subject to E-completeness and the violation, all other relation values are arbitrary. Thus legality is exactly the satisfaction of (6.2)–(6.5). No additional assumptions about the predicates or the binary and ternary relations are imposed.

Lemma 6.4. *The family $\mathcal{F}_*$ is finite and consists of irreducible structures of size three or four. The class $\mathcal{K}_* := \mathrm{Forb}_e(\mathcal{F}_*)$ has a countably infinite homogeneous Fraïssé limit K_*.*

Proof. The language and the listed domains are finite. Every forbidden structure is E-complete, hence irreducible. Free amalgamation follows from [13, Theorem 3.2]. The class is hereditary, has countably many isomorphism types, and has arbitrarily large members with every relation empty. Fraïssé's theorem [7] gives K_*. ∎

6.3. Encoding directed-graph homomorphisms

Let H be a nonempty finite loopless directed graph on $\{1, \dots, k\}$. Opposite arcs are permitted. A homomorphism preserves arcs and need not be injective. Write $J \to H$ when such a map exists, and put

$$Z_H = \{(i, j) \in \{1, \dots, k\}^2 : (i, j) \notin E(H)\}.$$

In particular, $(i, i) \in Z_H$ for every i.

Define the boundary A_H on the disjoint union

$$\{a_0, \dots, a_{k+1}\} \,\dot\cup\, \{b_{ij} : (i, j) \in Z_H\}$$

by the following tables:

$$\begin{aligned}
O^{A_H} &= A_H, & V^{A_H} &= \varnothing,\\
\mathsf{Path}^{A_H} &= \{a_0, \dots, a_{k+1}\}, & \mathsf{Test}^{A_H} &= \{b_{ij} : (i, j) \in Z_H\},\\
\mathsf{First}^{A_H} &= \{a_0\}, & \mathsf{Last}^{A_H} &= \{a_{k+1}\},\\
E^{A_H} &= \{(x, y) : x, y \in A_H,\ x \neq y\}, & N^{A_H} &= \{(a_l, a_{l+1}) : 0 \le l \le k\},\\
L^{A_H} &= \{(b_{ij}, a_i) : (i, j) \in Z_H\}, & R^{A_H} &= \{(b_{ij}, a_j) : (i, j) \in Z_H\}.
\end{aligned}$$

All other relation values are false. Let δ be the singleton on which V alone holds. The type t_H over A_H gives $*$ singleton type δ, sets $E(*, a) = E(a, *) = 1$ for every $a \in A_H$, and makes every other mixed relation value false. Both A_H and t_H are legal, since each forbidden condition requires at least two V-vertices. Embed A_H into K_* and put

$$P_H = P(t_H), \qquad C_H = \mathrm{Age}(K_* \restriction P_H).$$

For a finite L_*-structure B and $h \in B$, define a loopless directed graph $J_h(B)$ by

$$D_h(B) = \{v \in B \setminus \{h\} : Q(h, v) \wedge E(h, v) \wedge E(v, h)\}, \tag{6.6}$$

$$V(J_h(B)) = D_h(B),$$

$$E(J_h(B)) = \{(v, w) \in D_h(B)^2 : v \neq w,\ E(v, w) \wedge E(w, v) \wedge G(v, w)\}. \tag{6.7}$$

The empty graph has a homomorphism into every H.

Proposition 6.5. *For every finite L_*-structure B,*

$$B \in C_H \iff \big(\forall x \in B\ B \restriction \{x\} \cong \delta\big) \text{ and } \big(\forall h \in B\ J_h(B) \to H\big). \tag{6.8}$$

No E-completeness assumption on B is required.

Proof. The extension property of K_* gives the following criterion:

$$\begin{aligned} B \in C_H \iff & \text{ there is } Y \in \mathcal{K}_* \text{ on } A_H \dot\cup B \text{ with} \\ & Y \restriction B = B, \qquad Y \restriction (A_H \cup \{x\}) = t_H \quad (x \in B), \end{aligned} \tag{6.9}$$

where the last equality sends $*$ to x.

Assume such a Y exists. Fix $h \in B$ and $v \in D_h(B)$. By (6.2) there is a unique $c_l(h,v) \in \{0,1,2\}$ such that $S_{c_l(h,v)}(a_l,h,v)$ holds. Equations (6.2) and (6.3) give

$$c_0(h,v) = 0, \qquad c_{k+1}(h,v) = 2, \qquad (c_l(h,v), c_{l+1}(h,v)) \in \mathcal{T}.$$

Consequently there is a unique $f_h(v) \in \{1,\dots,k\}$ with

$$c_l(h,v) = \begin{cases} 0 & l < f_h(v), \\ 1 & l = f_h(v), \\ 2 & l > f_h(v). \end{cases} \tag{6.10}$$

For every $(i,j) \in Z_H$, equation (6.4) yields

$$U_L(b_{ij},h,v) \iff f_h(v) = i, \qquad U_R(b_{ij},h,v) \iff f_h(v) = j. \tag{6.11}$$

All these applications are on E-complete sets: the boundary is E-complete, every old–new pair has both E-directions, and $v \in D_h(B)$ supplies the remaining pair.

If $(v,w) \in E(J_h(B))$ and $(f_h(v), f_h(w)) \in Z_H$, take $b = b_{f_h(v),f_h(w)}$. The four points b,h,v,w satisfy the hypotheses of (6.5), whereas both prohibited bits hold by (6.11). Hence $(f_h(v), f_h(w)) \in E(H)$. Thus $f_h : J_h(B) \to H$ is a homomorphism, proving necessity.

Conversely, assume the right side of (6.8) and choose $f_h : J_h(B) \to H$ for each $h \in B$. Retain the whole tables on A_H and B. Set all old–new E-pairs true and every other mixed binary value false. For $v \in D_h(B)$ prescribe

$$\begin{aligned} S_0(a_l,h,v) &\iff l < f_h(v), \\ S_1(a_l,h,v) &\iff l = f_h(v), \\ S_2(a_l,h,v) &\iff l > f_h(v). \end{aligned} \tag{6.12}$$

$$\begin{aligned} U_L(b_{ij},h,v) &\iff f_h(v) = i, \\ U_R(b_{ij},h,v) &\iff f_h(v) = j. \end{aligned} \tag{6.13}$$

Every other mixed ternary value is false. These prescriptions concern distinct ordered tuples. In particular, the choices for hubs h and v do not conflict. Every prescribed ternary tuple uses two distinct new vertices, so each new point still has type t_H over A_H.

It remains to check legality. The unary tables force every O-vertex of a potential forbidden configuration into A_H and every V-vertex into B. The E, Q hypotheses force each relevant v to belong to $D_h(B)$. Exactly one of $l < f_h(v)$, $l = f_h(v)$ and $l > f_h(v)$ holds, so (6.12) assigns exactly one state. Since $1 \le f_h(v) \le k$, the endpoint states are 0 and 2, verifying (6.2). Adjacent states belong to $\mathcal{T}$, which gives (6.3). The definitions also give

$$L(b_{ij}, a) \Rightarrow a = a_i, \qquad R(b_{ij}, a) \Rightarrow a = a_j;$$

together with (6.12)–(6.13), these implications give (6.4). Finally, a violation of (6.5) at b_{ij}, h, v, w would give

$$(v, w) \in E(J_h(B)), \qquad f_h(v) = i, \quad f_h(w) = j, \qquad (i, j) \in Z_H,$$

contradicting that f_h is a homomorphism. Thus the constructed Y is legal, and (6.9) proves sufficiency. ∎

Corollary 6.6. *For finite nonempty loopless directed graphs H, H',*

$$C_H \subseteq C_{H'} \quad \Longleftrightarrow \quad H \to H'. \tag{6.14}$$

Proof. The reverse implication follows by composing the homomorphisms in (6.8). For the forward implication, form B_H on $\{h\} \dot\cup V(H)$ by giving all points type δ, making E complete, and setting

$$Q^{B_H} = \{(h, v) : v \in H\}, \qquad G^{B_H} = E(H).$$

All other values are false. Then $J_h(B_H) = H$ and $J_v(B_H) = \varnothing$ for $v \neq h$. Hence $B_H \in C_H$, and $B_H \in C_{H'}$ implies $H \to H'$. ∎

6.4. *Infinite degrees*

Proof of Theorems 0.5 and 6.1. For $m \ge 2$, let D_m be the directed cycle on $\mathbb{Z}/m\mathbb{Z}$ with arcs $i \to i + 1$. A homomorphism $g : D_p \to D_q$ necessarily satisfies $g(i) = g(0) + i \pmod q$. Therefore

$$D_p \to D_q \quad \Longleftrightarrow \quad q \mid p.$$

The ages C_{D_p}, with p ranging over the primes, are pairwise incomparable by Corollary 6.6. Their types t_{D_p} all have singleton type δ in the same structure K_*. For every integer $q \ge 2$, Proposition 6.3 gives a q-colouring of $(K_*)_\delta$ using all q colours in every copy of K_*. Thus

$$T_{\mathrm{emb}}(\delta, K_*) = T_{\mathrm{copy}}(\delta, K_*) = \infty.$$

Let $A \in \mathrm{Age}(K_*)$ have $k \ge 1$ vertices of type δ, and fix one such vertex $a \in A$. In every copy $K' \cong K_*$, each vertex of type δ is the image of a under an embedding of A into K'. For the preceding q-colouring χ_q, the colouring

$$e \longmapsto \chi_q(e(a)), \qquad e \in \mathrm{Emb}(A, K_*),$$

uses all q colours on every copy, proving $T_{\mathrm{emb}}(A, K_*) = \infty$.

For copies of A, use the finite colouring

$$B \longmapsto \{\chi_q(x) : x \in B \cap (K_*)_\delta\}.$$

Each value has size at most k, and the union of its values over any copy of K_* contains all q colours. A bound of t values would imply $q \leq kt$. Choosing $q = kt + 1$ excludes every finite t. ∎

Corollary 6.7. *The structure K_* has no precompact relational big Ramsey structure. More strongly, it has no precompact relational expansion in which every realized expanded singleton has big Ramsey degree one.*

Proof. Suppose such an expansion has m singleton types above δ. Take the persistent colouring χ_{m+1} above. Successively apply singleton degree one to the pullback of this colouring on each expanded singleton type. Composing the resulting self-embeddings gives an expanded copy on which each of the m types is monochromatic. Its reduct is a copy of K_* with at most m colours on the δ-vertices, a contradiction. ∎

Remark 6.8. The counterexample uses several unary, binary and ternary relations; its ternary relations need not be symmetric. It is not a counterexample in the language of a single symmetric ternary relation. In particular, the proof does not settle the question about the homogeneous 3-uniform hypergraph omitting the irreducible four-vertex hypergraph with three edges. Nor do we assert that every irreducible forbidden family containing a four-vertex structure gives infinite degrees.

7. Further questions

The minimum in (5.7) is taken over all self-embeddings. A finite criterion for which collections of normal forms occur on a full generic copy would give an effective description of the exact degrees. The envelope bound (0.2) also leaves room for improvement.

The expansions constructed at finite stages are recurrent. The argument does not establish recurrence or the infinite Ramsey property for their union. Establishing these properties would strengthen Corollary 0.2 within the framework of [9, 16].

For finitely constrained free amalgamation classes, a general criterion for finite big Ramsey degrees must distinguish the binary results of [1,17] from Theorem 0.5. The counterexample uses several directed relations of different arities. It does not decide the corresponding question for a single symmetric ternary relation, including the homogeneous 3-uniform hypergraph omitting the four-vertex, three-edge irreducible hypergraph discussed in [8]. Nor does the present argument determine the least maximum forbidden size for a counterexample.

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